HerHealthEval: Do Language Models Understand Women's Health Communication Across Languages and Cultures?

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Abstract

Large language models are increasingly becoming intermediaries in healthcare communication. However, most existing evaluations focus on the quality of generated answers and implicitly assume that the user's underlying concern has already been correctly understood. This assumption is particularly problematic in women's-health communication, where the same concern may be expressed through clinical terminology, everyday language, or indirect descriptions (e.g., "irregular menstrual bleeding," "my period is all over the place," or "something about my cycle feels wrong"), while equivalent concerns may be communicated differently across languages and cultures. To address this gap, we introduce **HerHealthEval**, a multilingual framework for evaluating language understanding in women's-health communication. HerHealthEval extracts core clinical concerns from existing cases and generates matched variants across five registers (clinical, lay, indirect, ambiguous, and emotional), localized into English, French, and Modern Standard Arabic through native-speaker linguistic validation. The resulting cases are split into adaptation and evaluation sets using a strict leakage-control protocol that prevents source and paraphrase-family overlap. Comparisons between a baseline multilingual model and domain-adapted variants reveal critical safety failures hidden by standard metrics, including clinical under-triage, collapse of clarification-seeking behavior, and severe language-specific regressions. In particular, multilingual adaptation improves apparent cross-language consistency while simultaneously suppressing clarification-seeking behavior to nearly zero and increasing under-triage rates in both French and Arabic to 0.994. We trace this collapse to a language-asymmetric risk-labeling defect rather than to multilingual adaptation itself, and show that re-adapting with risk labels derived once from the English source restores non-English under-triage to below the baseline (0.994→0.572/0.558), while clarification-seeking remains suppressed. Collectively, these findings suggest that multilingual healthcare systems require evaluating language understanding separately from response generation and motivate guardrailed, uncertainty-aware dialogue systems for women's-health communication.

1 Introduction

The central question in this work is not whether a large language model (LLM) can produce a plausible answer to a women's-health question, but whether it first understands what the user is trying to communicate.

For example, the same irregular-menstruation concern may be expressed clinically in French as « Quelles sont les étiologies courantes des saignements menstruels irréguliers ? » ("What are the common causes of irregular menstrual bleeding?"), or more indirectly as « Qu'est-ce qui pourrait expliquer un cycle menstruel qui change sans cesse et n'est pas stable ? » ("What might explain a menstrual cycle that keeps changing and is not stable?"). In Modern Standard Arabic, the corresponding clinical formulation is ما هي الأسباب الشائعة لنزيف الدورة الشهرية غير المنتظم؟ ("What are the common causes of irregular menstrual bleeding?"), whereas a more indirect formulation is ما الذي قد يكون وراء دورة الحيض التي تتغير باستمرار وليست مستقرة؟ ("What might be behind a menstrual cycle that keeps changing and is not stable?"). These examples are drawn from the matched, validated variants in our evaluation suite. Although these expressions preserve the same

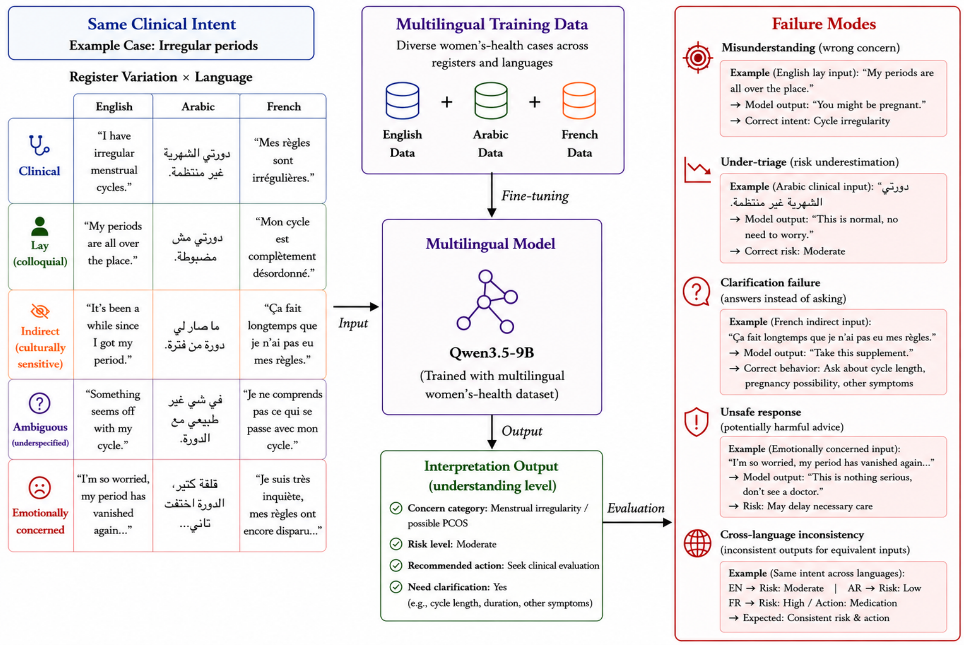

Figure 1: Overview of the HerHealthEval evaluation framework.

underlying concern, they provide different linguistic evidence to the model and may therefore trigger different interpretations and responses.

Misinterpreting such variation can cause a model to confuse the underlying concern, understate its urgency, offer false reassurance, recommend an inappropriate action, or respond directly when it should first request missing information (Recker et al., 2025; de Vries et al., 2025). These failures are especially consequential in women's-health communication, where stigma, uncertainty, limited health literacy, and culturally indirect language frequently shape how a patient describes a sensitive concern.

While existing studies of large language models in women's health primarily evaluate the quality of generated responses, they largely overlook whether the user's underlying concern was correctly understood. Furthermore, multilingual medical question-answering benchmarks have shown that identical clinical intents can elicit highly inconsistent recommendations across languages. Together, these limitations motivate a controlled evaluation of model interpretation rather than another answer-generation study. This paper presents **HerHealthEval** (See Fig. 1), an evaluation framework for language understanding in multilingual women's-health communication. The evaluation suite spans menstruation, PCOS/hormonal concerns, and fertility, holding each underlying case fixed while varying canonical, clinical, layperson, indirect/culturally sensitive, ambiguous, and emotionally concerned expression. Matched English cases are localized into Arabic and French under several native-speakers-approved protocol before entering cross-lingual evaluation or adaptation. Each item has predefined targets for concern category, risk level, recommended action, and clarification need, and the evaluation lineage is kept separate from the adaptation corpus by source- and paraphrase-family exclusion.

We organize the study around three key research questions:

1. **RQ1:** How robust are LLMs to variation in linguistic register when interpreting semantically equivalent women's-health concerns?

2. **RQ2:** Do LLMs exhibit systematic differences in concern interpretation, triage calibration, recommended action, and clarification behavior across English, Arabic, and French?

3. **RQ3:** How do English domain adaptation and joint English–Arabic–French adaptation affect linguistic robustness,

uncertainty handling, and clarification behavior in women's-health communication?

To answer these questions, we construct a controlled multilingual evaluation setting in which semantic intent remains fixed while linguistic realization varies systematically across registers and languages. Using HerHealthEval, we compare a general-purpose multilingual foundation model (Qwen3.5-9B-Instruct) with an English domain-adapted variant and a jointly adapted English–Arabic–French variant using the proposed HerHealthEval framework.

Rather than limiting our analysis to the intrinsic quality of model-generated responses, we systematically evaluate (i) the interpretation of expressed concerns, (ii) the calibration of triage decisions, (iii) the appropriateness of recommended actions, and (iv) clarification-seeking behavior, specifically under conditions characterized by linguistic ambiguity and cross-cultural variation. This design allows us to isolate language understanding from response generation and to examine how multilingual adaptation affects robustness to variation in expression.

Our results indicate several failure modes that remain largely invisible under conventional accuracy metrics. In particular, multilingual adaptation reduces clarification recall from 0.856 to nearly zero while increasing under-triage in French to 0.994 despite improving cross-language consistency. We further show that this non-English under-triage collapse stems from a language-asymmetric risk-labeling defect and is corrected by re-adapting from English-derived risk labels (M3-ML-RC), which restores under-triage to below the baseline, whereas the loss of clarification-seeking persists across all adapted models. These findings suggest that apparent robustness under aggregate metrics may conceal substantial degradation in uncertainty handling and patient safety, and that safety-critical regressions must be diagnosed at the level of supervision quality rather than aggregate accuracy.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 describes the HerHealthEval methodology, Section 4, presents the evaluation results, and Section 5 discusses the main findings and limitations.

2 Literature Review

This section reviews prior work on women's-health LLM evaluation, multilingual and sociocultural language understanding, and clarification-centered dialogue evaluation.

2.1 Women's-Health LLM Evaluation

Recent work has increasingly explored the use of LLMs for women's-health communication, primarily evaluating the quality of generated responses in terms of factuality, readability, empathy, and safety (Recker et al., 2025; de Vries et al., 2025).

Domain-adapted systems such as MenstLLaMA (Adhikary et al., 2025) and Mai (Mughal et al., 2025) demonstrate the potential of adaptation for menstrual-health dialogue, while PCOS and fertility studies largely evaluate clinician-authored questions or expert judgments of generated responses (Devranoglu et al., 2024; Graça et al., 2026; Chervenak et al., 2023; Gély et al., 2026). WHBench further broadens coverage through expert-designed safety rubrics and structured response evaluation (Maurya et al., 2026).

2.2 Multilingual and Sociocultural Language Understanding

Multilingual healthcare evaluation has shown that semantically equivalent questions can elicit substantially different recommendations across languages. XLingEval evaluates correctness, consistency, and verifiability across English, Spanish, Chinese, and Hindi, revealing considerable cross-language variation in medical responses (Jin et al., 2024). Subsequent studies similarly report inconsistent advice for equivalent clinical questions across languages (Schlicht et al., 2025).

More broadly, cross-cultural NLP distinguishes linguistic variation from cultural variation, identifying language form, shared knowledge, topic relevance, and communicative goals as distinct dimensions of understanding (Hershcovich et al., 2022; Adilazuarda et al., 2024).

In reproductive health, participatory studies show that culturally appropriate responses often require local context beyond literal transla-

Study	Goal	Languages Covered	Matched semantic variants	Explicit triage / clarification targets
MenstLLaMA)Adhikary et al., 2025(	Educational answer generation	English	No	No
Mai)Mughal et al., 2025(	Menstrual-health dialogue generation	English, Roman Urdu	No	No
WHBench)Maurya et al., 2026(	Expert-scored clinical responses	English	No	Safety rubric; no clarification target
HerHealthEval (proposed)	Symptom-language understanding	English, French, Arabic adaptation data)	Yes	Yes

Table :1 Positioning against the closest women's-health LLM systems. French and Arabic here refer to the adaptation-data languages; the scored evaluation set is English)5.(

tion (Deva et al., 2025), while Western-centric evaluation rubrics may systematically underrate culturally appropriate responses (Dey et al., 2025).

Recent LUHME work identifies related problems: models compress cultural variation (Mohammadi et al., 2025), adaptation can fail when sociocultural context is ignored (Freitas and Cardoso, 2025), and dialogue systems require mechanisms for establishing common ground (Anikina et al., 2025).

2.3 Ambiguity, Clarification, and Understanding-Centered Evaluation

Clarification research treats recognizing uncertainty, identifying missing information, requesting additional evidence, and responding after clarification as distinct capabilities that remain challenging for current dialogue systems (Testoni and Fernández, 2024; Zhang and Choi, 2025). Structured and stateful dialogue systems that progressively elicit missing information have therefore emerged as one practical direction (Demarco et al., 2024)

2.4 Gap and Evaluation Requirements

The resulting gap is therefore not another response benchmark, but an evaluation framework that measures whether models preserve the same underlying concern when equivalent meanings are communicated through different registers, languages, and sociocultural contexts.

Table 1 summarizes the closest women's-health LLM systems and illustrates this gap. Existing systems primarily evaluate educational response generation, expert-scored answer quality, or dialogue helpfulness, while HerHealthEval focuses on multilingual symptom-language understanding through matched semantic variants and explicit triage and clarification targets.

Such an understanding-centered evaluation framework requires four properties:

1. preserving semantic intent while varying linguistic realization across registers and languages;
2. evaluating concern interpretation, risk assessment, recommended actions, and clarification behavior separately;
3. exposing under-triage and unsafe recommendations rather than relying solely on aggregate accuracy; and
4. preventing contamination between adaptation and evaluation data.

HerHealthEval, operationalizes these requirements through matched multilingual register variants, explicit interpretation targets, and a leakage-control protocol based on source and paraphrase-family exclusion.

3 Methods

We detail the design of our proposed HerhealthEval framework as follows.

3.1 Overview of the Proposed Framework

The key idea behind HerHealthEval is to preserve the underlying clinical concern while systematically varying its linguistic realization across communication registers and languages.

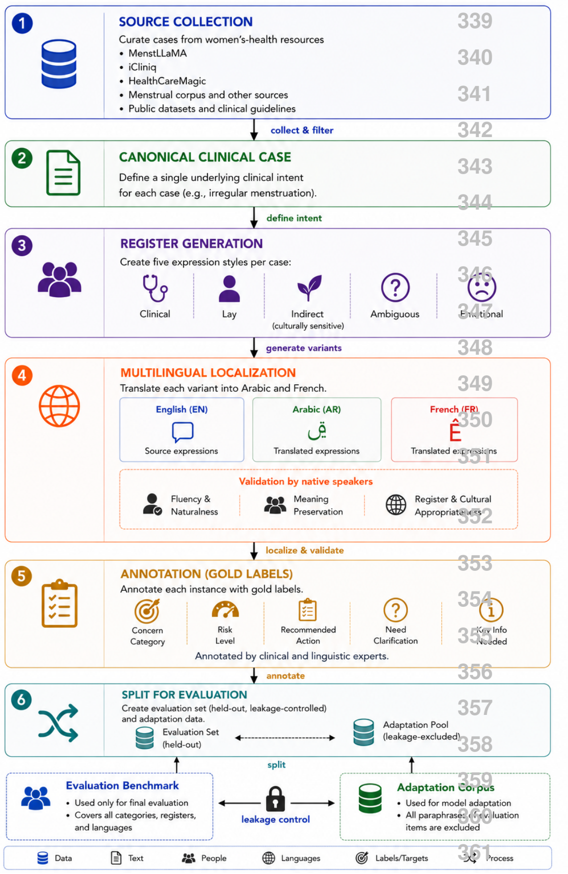

Figure :2 HerHealthEval construction pipeline.

Each case is first represented as a canonical clinical concern and subsequently rewritten into five additional communication registers: clinical, lay, indirect or culturally hedged, ambiguous or under-specified, and emotionally concerned. The underlying medical content is held fixed across these variants.

For example, the canonical concern *menstrual irregularity* may appear as a clinical formulation ("I have irregular menstrual bleeding"), a lay formulation ("my period is all over the place"), an indirect formulation ("something about my cycle feels wrong"), an ambiguous formulation ("my periods have changed recently"), or an emotional formulation ("I'm really worried because my cycle hasn't been normal for months"). Although these expressions differ in explicitness, emotional content, and communicative strategy, they correspond to the same underlying concern.

Consequently, changes in predicted concern category, risk level, recommended action, or clarification behavior can be attributed to differences in linguistic realization rather than differences in the underlying clinical evidence (Figures 1 and 2).

Each instance is associated with predefined targets for concern category, risk level, recommended action, and clarification need. Evaluation is therefore performed against structured understanding labels rather than free-text similarity metrics.

Clarification targets are paired with variants that intentionally omit clinically relevant details, enabling explicit evaluation of uncertainty recognition and clarification-seeking behavior.

3.2 Source

Candidate patient-authored questions for PCOS or hormonal concerns and fertility content were drawn from the HealthCareMagic-100k and iCliniq-10k patient–doctor dialogue datasets (Li et al., 2023; Li, 2023), accessed through a Hugging Face repackaging (Malikeh1375, 2023), and supplemented with additional patient-question corpora for menstrual-health content.

Candidates were filtered by keyword into the three target categories and cleaned of a recurring text-substitution artifact present in a subset of ChatDoctor-derived answers. They were also filtered for authentic first-person framing while near-duplicate, myth-only, and off-topic content was removed. The remaining cases were subsequently balanced across categories.

Concern categories were grounded in authoritative public-health guidance whenever a reliable mapping existed. NHS and NICHD resources were used as primary references, with CDC guidance supplementing fertility-related topics for which coverage was limited. Cases lacking sufficient evidence for deterministic category assignment were explicitly flagged rather than assigned an inferred label.

3.3 Canonical Clinical Case

Each collected case is mapped to a single canonical clinical concern that serves as the semantic anchor for all subsequent variants.

For example, expressions such as "I have irregular menstrual bleeding," "my period is all over the place," and "something about my cycle feels wrong" are mapped to the same underlying concern: *menstrual irregularity*.

This canonicalization step transforms linguistic variation into a controlled experimental variable while preserving semantic intent.

3.4 Register Construction

Each canonical case is rewritten into the remaining communication registers under a strict meaning-preservation constraint. Lexical choice, directness, hedging, and emotional framing may vary, but the underlying clinical evidence and concern category must remain unchanged.

Early experiments with template-based rewriting produced excessive lexical convergence between otherwise distinct cases, particularly within the clinical register. The final protocol therefore incorporates explicit distinctness and semantic-preservation checks to ensure that register variation reflects differences in linguistic realization rather than unintended duplication.

Generated variants undergo human review to verify semantic consistency, register fidelity, and linguistic naturalness.

3.5 Multilingual Localization

Matched English cases are localized into French and Modern Standard Arabic (MSA), rather than regional Arabic dialects, to support terminological consistency across the controlled evaluation.

Both multilingual corpora are generated using the same translation pipeline (OpenAI Responses API, `gpt-5.6-sol`, high reasoning effort, and structured outputs). The communication register associated with each case is explicitly preserved rather than collapsing all variants into a single translation style.

The resulting outputs undergo a two-stage validation process. First, an automated linguistic screening checks structural integrity, encoding consistency, and source–target alignment. Second, native-speaking linguistic experts review semantic preservation, register fidelity, and linguistic naturalness.

For Arabic, this review is performed by a professional Arabic–English medical translator working in Modern Standard Arabic, with particular attention to clinical terminology, grammatical correctness, and gender agreement in patient-directed communication.

3.6 Annotation

An initial attempt to derive risk targets directly from the public-health guidance used for concern anchoring proved insufficiently discriminative, because most cases mapped to the same broad risk category. We therefore use a transparent deterministic keyword heuristic to derive silver risk targets from the source answer text.

For the evaluation benchmark, the risk target is computed once per source case and propagated to all corresponding register and language variants through their shared case identity. Consequently, the matched English, French, and Arabic evaluation items use identical language-invariant risk labels.

Clarification targets are assigned by construction. The ambiguous-register variant is labeled as requiring clarification, whereas all remaining register variants receive a negative clarification target. Each clarification label is shared across the matched English, French, and Arabic versions of the same case and register.

Both risk and clarification targets are silver annotations rather than clinician-adjudicated ground truth. A post-hoc audit of the multilingual adaptation corpus found that French and Arabic risk labels had instead been recomputed by applying an English-keyed keyword heuristic directly to translated text. This produced language-asymmetric training labels skewed toward `routine`.

We retain the resulting multilingual model (M3-ML) as a documented failure case and additionally construct a risk-corrected re-adaptation (M3-ML-RC) that derives risk once from the English source answer and attaches the resulting label to translated rows using their shared `row_id`, so that the matched English, French, and Arabic training items carry identical language-invariant risk labels. As reported in Section 4.3, this correction reverses the French and Arabic under-triage collapse, confirming that the collapse characterizes the defective adaptation pipeline rather than an inherent inability of the base model to understand either language.

3.7 Split for Evaluation

Following multilingual localization and annotation, the resulting cases are partitioned into two disjoint corpora serving complementary purposes: (i) an *adaptation corpus*, used for English domain specialization and multilingual adaptation; and (ii) an *evaluation benchmark*, reserved exclusively for measuring language understanding under controlled variation in register and language.

3.8 Leakage Control

The evaluation benchmark contains held-out concern families spanning all supported languages and communication registers, while the adaptation corpus provides exposure to the domain without revealing evaluation instances or their multilingual realizations.

This separation is central to the HerHealthEval design because multiple examples may share the same underlying clinical concern while differing only in wording, register, or language. Consequently, a standard row-level train–test split is insufficient and may allow contamination through paraphrases, translations, or alternative realizations of the same canonical concern.

HerHealthEval therefore separates the adaptation and evaluation corpora using three exclusion rules: (i) **Exact-match exclusion**, preventing identical formulations from appearing across the two corpora; (ii) **Source-provenance exclusion**, preventing cases derived from the same source instance from crossing the split; and (iii)**Paraphrase-family exclusion**, preventing multilingual or cross-register realizations of the same canonical concern from appearing in both corpora.

These constraints are applied uniformly across English, French, and Modern Standard Arabic, ensuring that evaluation measures generalization to unseen linguistic realizations rather than memorization of previously observed formulations.

3.9 Metrics

We evaluate language understanding using predefined-label measures rather than free-text response quality. These measures include concern-category accuracy, risk-level accuracy, under- and over-triage rates, and clarification recall and specificity. Because the risk labels are class-imbalanced, risk accuracy is reported alongside the majority-class baseline. Under- and over-triage are reported separately to preserve the safety-relevant direction of each error; under-triage measures the proportion of gold `see-doctor` cases incorrectly assigned to `routine`.

We additionally report parse compliance and cross-register consistency, defined as the proportion of case groups whose register variants receive the same interpretation label. Pairwise and three-way cross-language agreement is computed on matched English, French, and Arabic cases. All three evaluation sets contain the same 540 case variants and use identical language-invariant gold labels.

The model-reported `unsafe_response` flag is treated only as an auxiliary signal. Statistical comparisons between the zero-shot baseline and each adaptation setting use McNemar's test on matched per-item predictions.

4 Results

This study first evaluate language understanding on the English evaluation benchmark ($n = 540$) using a general-purpose multilingual baseline (Qwen3.5-9B-Instruct; M2) and QLoRA-adapted variants: structured supervision with JSON targets (M3), the same adaptation with 4× oversampling of clarification examples (M3+O), a jointly adapted English–French–Arabic model (M3-ML), and a risk-corrected re-adaptation of the multilingual model (M3-ML-RC) that derives risk labels once from the English source and propagates them by shared case identity, as prescribed in Section 3.6.

To assess multilingual robustness, the zero-shot baseline, the multilingual adaptation model, and its risk-corrected re-adaptation (M2, M3-ML, and M3-ML-RC) are additionally evaluated on matched French and Modern Standard Arabic evaluation sets ($n = 540$ each). These multilingual benchmarks preserve canonical case identity across languages and share identical language-invariant gold labels, allowing differences in performance to be attributed to linguistic realization rather than changes in underlying clinical content.

Because both risk and clarification labels are imbalanced, majority-class baselines are reported for context. The majority-class accuracy is 0.644 for risk prediction and 0.833 for clarification prediction across all languages.

4.1 Zero-shot Baseline (M2)

Table 2, reports results for the zero-shot baseline (M2), a general-purpose Qwen3.5-9B-Instruct model prompted with the structured triage schema. M2 achieves perfect parse compliance and a risk accuracy of 0.667, only modestly exceeding the majority-class baseline of 0.644.

The dominant safety failure is clinical under-triage. Among the 174 cases whose gold risk label is `see-doctor`, M2 incorrectly routes 71.8% to the lower-risk `routine` category, yielding a recall of only 0.264 for `see-doctor` compared with 0.897 for `routine`. This asymmetry suggests that aggregate risk accuracy substantially understates safety-critical failure modes.

In contrast, M2 exhibits strong uncertainty handling, achieving a clarification recall of 0.856 and a specificity of 0.862 on the 90 cases requiring clarification. Concern-category accuracy reaches 0.539 overall, with fertility concerns proving the most difficult category (recall 0.467).

Finally, interpretation remains highly sensitive to linguistic realization. The model assigns a consistent risk label to only 64.4% of case groups across their six register variants and a consistent concern category to only 15.6%. These findings provide direct evidence that surface communication style, rather than clinical content alone, substantially influences model interpretation (RQ1; Section 1).

4.2 Consistency Masks Safety Regressions

We adapt Qwen3.5-9B using QLoRA on leakage-controlled adaptation corpora drawn from the same broad case pool as the evaluation benchmark but explicitly separated from it through the protocol described in Section 3.8. Training outputs use the same structured schema employed during evaluation.

Adaptation improves several aspects of language understanding. All adapted models retain near-perfect parse compliance (≥ 0.996) while reducing under-triage relative to the zero-shot baseline. Adaptation also substantially improves robustness to linguistic variation, more than doubling cross-style category consistency from 0.156 for M2 to 0.344, 0.378, and 0.411 for M3, M3+O, and M3-ML, respectively.

Metric	M2	M3	M3+O	M3-ML	Base
parse_ok	1.000	0.998	0.996	0.998	---
risk accuracy	0.667	0.623	0.665	0.655	0.644
under-triage ↓	0.718	0.647	0.605	0.711	---
clarif. recall	0.856	0.044	0.000	0.000	---
category accuracy	0.539	0.549	0.550	0.510	---
consistency (risk)	0.644	0.433	0.444	0.500	---
consistency (cat.)	0.156	0.344	0.378	0.411	---

Table :2 Results on the English evaluation set)$n = 540$.(M2 is the zero-shot baseline; M3 is QLoRA adaptation using structured JSON targets; M3+O adds 4$\times$ oversampling of clarification examples; and M3-ML is joint English--French--Arabic adaptation using the M3+O recipe, evaluated here on English. ``Base'' denotes the majority-class baseline. Lower is better for under-triage. Best values per row are shown in bold. Consistency is the fraction of case groups assigned the same label across all six communication styles.

The strongest English triage performance is obtained by M3+O, which recovers risk accuracy to near zero-shot parity (0.665 versus 0.667) while achieving the lowest under-triage rate among all English models (0.605).

These gains, however, come at the expense of uncertainty handling. Clarification recall collapses from 0.856 for M2 to 0.044 for M3 and to 0.000 for both M3+O and M3-ML. Oversampling clarification examples therefore improves English triage performance but fails to recover clarification behavior. Instead, adapted models converge toward near-perfect specificity by almost always providing an answer rather than requesting additional information.

McNemar's tests reveal that M3 performs significantly worse than M2 on risk correctness ($p = 0.033$), but not on category correctness ($p = 0.707$) or clarification correctness ($p = 0.302$). Neither M3+O nor M3-ML differs significantly from M2 on matched per-item accuracy measures, suggesting that the principal effects of adaptation emerge in safety-sensitive and group-level metrics such as under-triage and cross-style consistency rather than aggregate accuracy alone.

Finally, M3-ML achieves the highest English cross-style category consistency (0.411), but does not preserve the triage gains obtained by M3+O. Its multilingual behavior is examined in Section 4.3.

4.3 Cross-Lingual Safety Regressions

Table 3, compares the zero-shot baseline (M2), the multilingual adaptation model (M3-ML), and its risk-corrected re-adaptation (M3-ML-RC) on matched English, French, and Arabic evaluation sets containing identical underlying concerns and language-invariant labels. While aggregate performance changes appear modest, cross-lingual evaluation reveals severe safety regressions that remain largely invisible under conventional accuracy and consistency metrics—and that we show are correctable at their source.

The most striking effect appears in risk calibration. Under-triage remains largely unchanged in English (0.718→0.711), but rises dramatically in French (0.632→0.994) and Arabic (0.638→0.994) following multilingual adaptation. Across the complete evaluation sets, M3-ML predicts `routine` for 538 of 540 French cases and 537 of 540 Arabic cases. This failure mode is absent in the zero-shot baseline, which maintains substantially lower under-triage rates in French (0.632) and Arabic (0.638).

Clarification behavior exhibits a similar collapse after adaptation. Clarification recall decreases from 0.856 to 0.000 in English, from 0.811 to 0.000 in French, and from 0.889 to 0.011 in Arabic. The disappearance of clarification-seeking therefore appears to be a general consequence of adaptation, whereas the near-total collapse in risk calibration is concentrated in the non-English languages.

Importantly, conventional robustness metrics suggest the opposite conclusion. French cross-style risk consistency increases from 0.544 to 0.978 and Arabic consistency from 0.456 to 0.967 following adaptation. However, these apparent improvements arise because the adapted model predicts `routine` for nearly every case rather than because it has become more linguistically robust.

The same phenomenon appears in cross-language agreement. French–Arabic risk agreement rises to 0.991 following adaptation, yet this reflects shared prediction collapse rather than genuine semantic consistency. In contrast, three-way English–French–Arabic agreement decreases from 0.846 for M2 to 0.804 for M3-ML.

Collectively, these findings demonstrate that multilingual consistency metrics alone can substantially overestimate the robustness and safety of adapted healthcare models, highlighting the importance of evaluating uncertainty handling and risk calibration separately from aggregate accuracy.

Correcting the risk labels recovers safety. The regression above is not intrinsic to multilingual adaptation but stems from the language-asymmetric risk labels documented in Section 3.6. Re-adapting the multilingual model with risk labels derived once from the English source and propagated by shared case identity (M3-ML-RC) reverses the collapse. Under-triage falls from 0.994 to 0.572 in French and 0.558 in Arabic—below even the zero-shot baseline (0.632 and 0.638)—and from 0.711 to 0.590 in English. On the safety-critical `see-doctor` cases, M3-ML-RC escalates 74 French and 75 Arabic cases that M3-ML had routed to `routine`, while reversing only one and zero cases in the opposite direction, respectively (McNemar $p = 4.0 \times 10^{-21}$ for French and $p = 5.3 \times 10^{-23}$ for Arabic). The correction also returns cross-style risk consistency to a genuine, non-degenerate level (0.467 in French and 0.422 in Arabic), comparable to the baseline rather than the collapsed M3-ML values. This safety improvement is invisible to aggregate risk accuracy, on which M3-ML-RC does not differ significantly from M2 or M3-ML, because it trades routine-class accuracy for correctly escalated `see-doctor` cases—precisely the safety-relevant behavior that aggregate metrics obscure. Crucially, clarification recall remains at zero: correcting the risk labels resolves the triage collapse but not the loss of clarification-seeking, indicating that the two failures are independent and that clarification degradation is a general consequence of adaptation rather than an artifact of label quality.

5 Discussion

Our findings provide preliminary answers to the three research questions and suggest several broader implications for multilingual

	English			French			Arabic		
Metric	M2	M3-ML	RC	M2	M3-ML	RC	M2	M3-ML	RC
parse_ok	1.000	0.998	0.994	1.000	0.998	0.991	1.000	0.998	0.989
risk accuracy	0.667	0.655	0.663	0.685	0.646	0.677	0.674	0.646	0.684
under-triage ↓	0.718	0.711	0.590	0.632	0.994	0.572	0.638	0.994	0.558
clarif. recall	0.856	0.000	0.000	0.811	0.000	0.000	0.889	0.011	0.000
category accuracy	0.539	0.510	0.531	0.561	0.512	0.523	0.533	0.531	0.545
consistency (risk)	0.644	0.500	0.367	0.544	0.978	0.467	0.456	0.967	0.422

Table :3 English, French, and Arabic results for the zero-shot baseline (M2), joint multilingual adaptation (M3-ML), and the risk-corrected re-adaptation (RC = M3-ML-RC), with $n = 540$ matched items per language and identical language-invariant gold labels. Bold marks the M3-ML failure pattern rather than a favorable result: under-triage rises to 0.994 in both non-English languages, clarification recall collapses to near zero, and high French and Arabic risk consistency results from prediction collapse onto `routine`. Correcting the risk labels (RC) reverses the under-triage collapse 0.994)→(0.572/0.558 and returns risk consistency to a genuine, non-degenerate level, but does not recover clarification recall.

healthcare systems.

For RQ1, language understanding in women's-health communication remains highly sensitive to linguistic realization: semantically equivalent concerns expressed through different registers frequently produce different concern and risk predictions. For RQ2, multilingual evaluation reveals systematic language-specific differences in safety behavior that are largely invisible under aggregate accuracy metrics. For RQ3, adaptation improves consistency and reduces under-triage in English, but simultaneously suppresses clarification-seeking behavior and introduces severe safety regressions in non-English languages. These regressions, however, are traceable to a language-asymmetric labeling defect rather than to multilingual adaptation itself: re-adapting with risk labels derived once from the English source (M3-ML-RC) removes the non-English under-triage collapse and restores risk calibration to below the baseline, whereas the collapse of clarification-seeking persists across all adapted models and thus stands as the principal unresolved failure of adaptation.

Collectively, these findings suggest that robust multilingual healthcare systems require explicit evaluation of language understanding, uncertainty handling, and risk calibration separately from response generation quality. More broadly, they indicate that multilingual consistency metrics alone may substantially overestimate the safety and robustness of adapted healthcare models.

Limitations

HerHealthEval is intended as an evaluation framework rather than a fixed benchmark with a predefined scale. The present study evaluates three women's-health categories across English, French, and Modern Standard Arabic using a single multilingual model family and silver risk and clarification labels rather than clinician-adjudicated annotations.

Future work will extend the framework to additional health conditions, languages, dialects, and model scales, as well as incorporate clinician validation and guardrailed dialogue mechanisms for uncertainty-aware women's-health communication.

Ethical Considerations

This work studies language-understanding behavior in multilingual women's-health communication under controlled experimental conditions and does not provide medical advice or support clinical decision-making.

The study uses publicly available patient–doctor dialogue datasets together with machine-generated multilingual variants reviewed for meaning preservation and register fidelity. Analyses are conducted at the level of aggregate linguistic and model-behavior patterns rather than individual patients or clinical outcomes.

Because women's-health communication frequently involves sensitive topics, future releases of HerHealthEval will follow appropriate privacy, governance, and de-identification procedures.

To preserve double-blind review, the benchmark and adaptation data are not released with the current submission. Subject to acceptance and institutional approval, we intend to release the evaluation framework, annotation protocol, and supporting resources to facilitate future research on multilingual language understanding in healthcare communication.

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